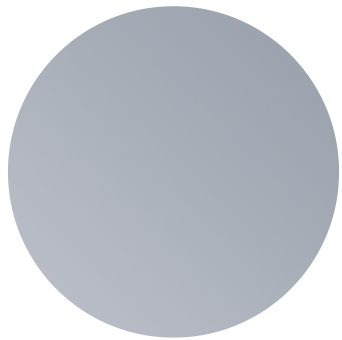




Data Driven Education Natural Language Processing (NLP) sub team

Presentation 1 – Fall 2019
09/18/2019

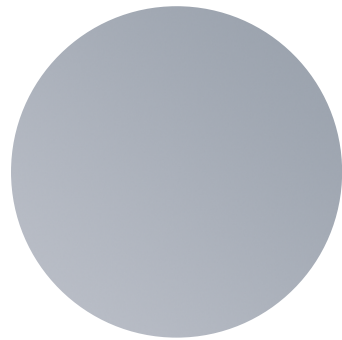
NLP SUB-TEAM



Jianin Jin

Data Scientist

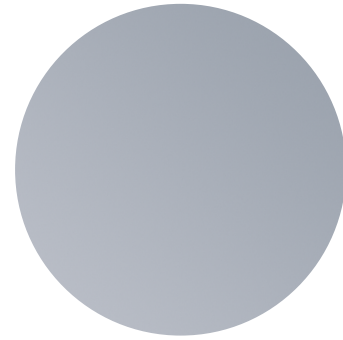
CS



Xinran Yu

Data Scientist

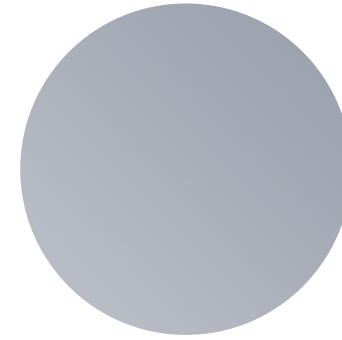
IE 2022



Atush Dhakal

Project Manager

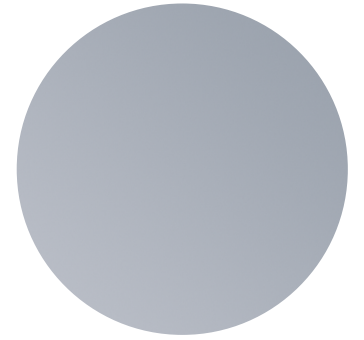
CS 2021



Lindsey Blackmore

Backend Developer

CS 2021



Zachary Abraham

Backend Developer

CS 2023

The Problem

Due to a large number of student queries (>5000 per semester) on Piazza, quick response/help from Professor and TA's is extremely challenging

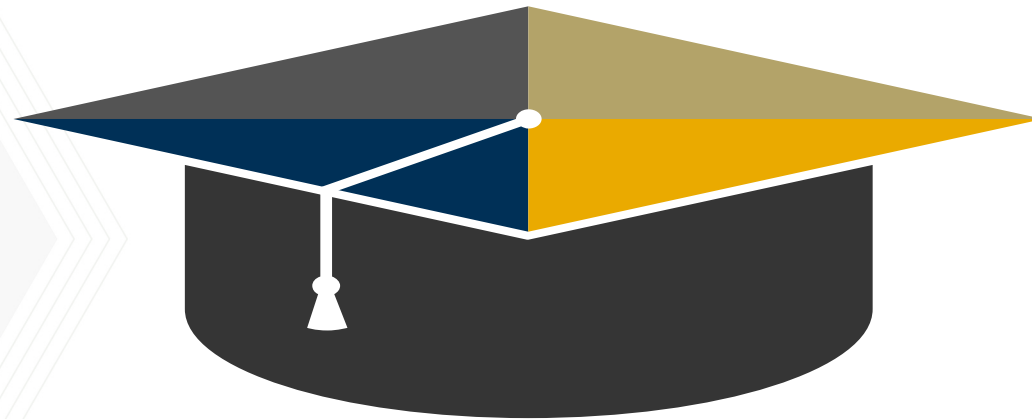


Large number of students are highly engaged through Piazza. Hence, it is important to quantify issues and the sentiment students express on Piazza

Potential Ideas

Piazza Analysis

Quantifying some of the most asked questions, topics, and keywords on Piazza



Personalized Resource Recommendation

Recommending personalized resources from Canvas /web to students based on their question, performance, and topics of interests

Semantic Analysis

Helping instructors and TA's with notification priority. Also quantifying students view on homework and tests

Piazza Analysis

- Quantifying some of the most asked questions, topics, and keywords on Piazza
- Assist in setting the agenda for plus session, recitation and test reviews

The goal of this project is to answer questions such as:

1. What are the leading chapter/topic/concept at which students are struggling?
2. How did students react to a particular assignment (includes sentiment analysis)?

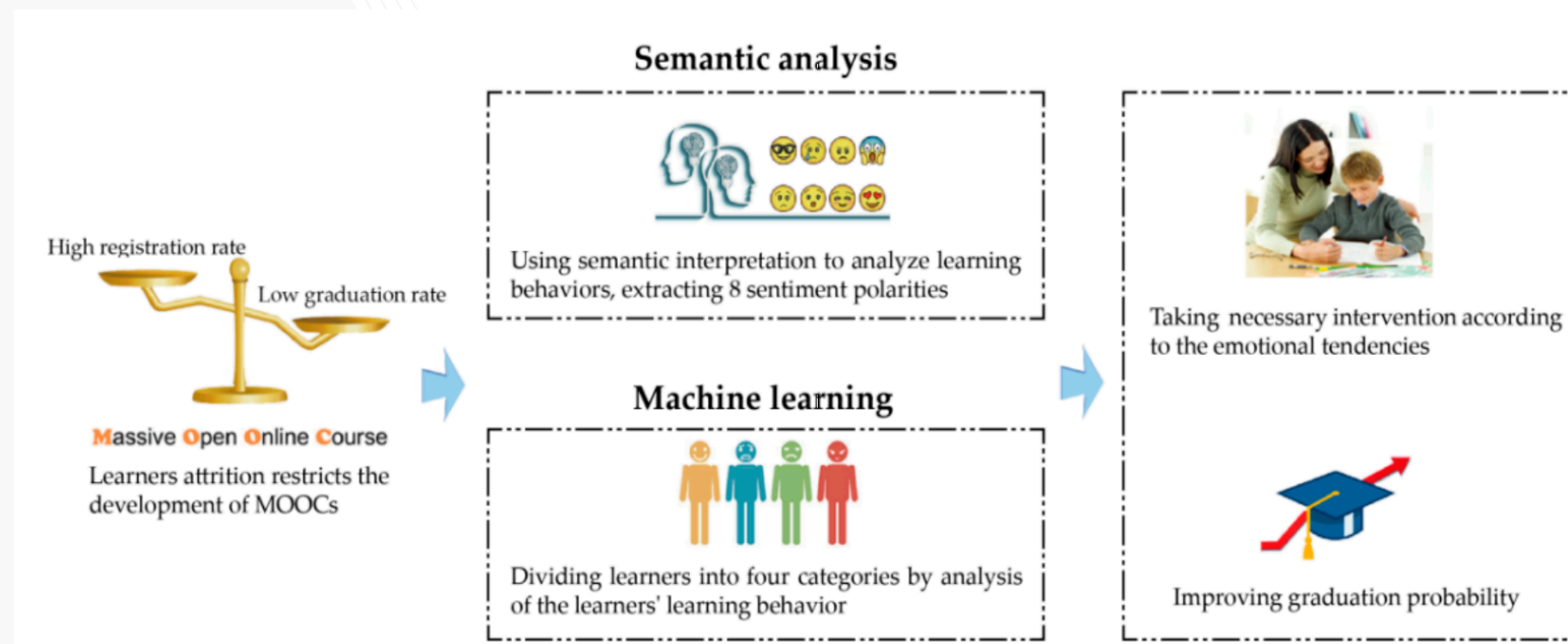
Personalized Resource Recommendation

- Recommending personalized resources from Canvas /web to students based on their question, performance, and topics of interests
- This tool will help instructor and TA's automate answer to questions such as:
 1. Where can I find homework 5?
 2. Where I can view the lecture slide for week 5?
 3. What does abstraction mean?

In each of these case, our tool will look for appropriate resources on Canvas or web (OERs) and answer students's queries

Semantic Analysis

- Provides quantitate and qualitative reports on students opinion regarding particular chapter, assignment, test and course in general
- Qualitative data may include polarity of the opinion(positive or negative), subject classification(types of sentence) and scope of the opinion(what does it relate to)
- Quantitative data may include “x amount of students felt....about test 4”
- Also useful in prioritizing notifications based upon time, subject, scope, and student performance
- Can be ultimately used to automate response



Source: Ling Wang, Gongliang Hu and Tiehua Zhou - Northeast Electric Power University

Our Progress

August and September

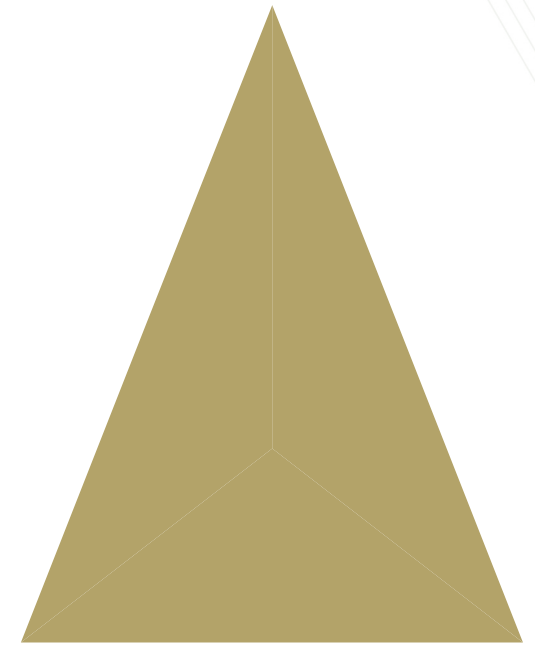
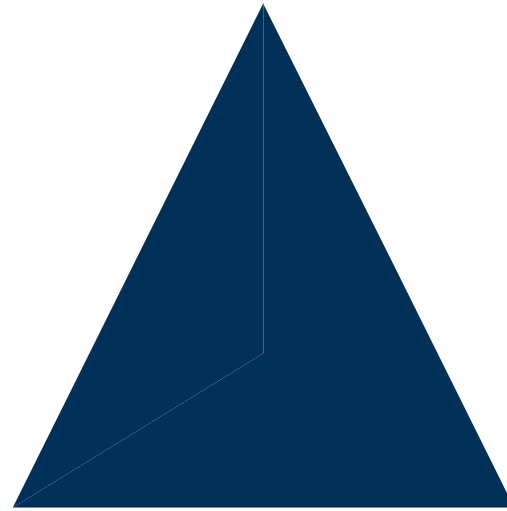
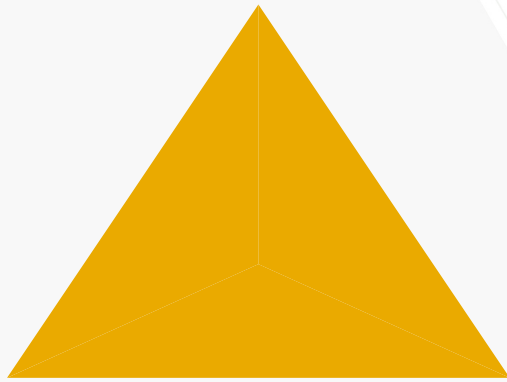
08/28	VIP Sub-teams announced
09/08	IRB certification completed
09/09	First VIP sub team meeting - project ideas discussion
09/11	Environment setup - Docker and GitHub completed
09/20	Second sub team meeting - semester preview, goals

SEPTEMBER

MON	TUE	WED	THU	FRI	SAT	SUN
26	27	28	29	30	31	1
2	3	4	5	6	7	8
9	10	11	12	13	14	15
16	17	18	19	20	21	22
23	24	25	26	27	28	29
30	1	2	3	4	5	6

Project timeline

23 September



- All team members will have access to data
- Second sub team meeting is completed
- Goals and idea are defined for the semester

Challenges



- Amount, type and constraints of the data set available
- Scaling the application university wide and beyond
- Timeframe and transition into next semester

Goals



Foundations

Lay strong foundations with progress that helps future sub teams scale the application

Scale

Ensure that the application is scalable beyond OMSA and benefits all undergraduate/graduate courses at GT and beyond

Impact

Develop tools that helps instructors understand their students better through real time reports which ultimately resulting in improved student academic performance

Questions